

# Construction Element Detection for Site Progress Verification

A cloud-only YOLOv8 pipeline for automated progress evidence

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## EXECUTIVE SUMMARY

Site progress in the Gulf is still recorded by photograph and narrative, with a two- to seven-day gap between capture and record. The photograph is archived; the information inside it is not. ISO 19650 asks for information delivered into a common data environment in a structured form, and a JPEG in a folder does not meet that bar.

This project trains a YOLOv8 detector on five construction classes — **steelbar**, **brick**, **pvcpipe**, **scaffold** and **excavator** — so that a site photograph yields a structured detection record: image, time, zone, class, confidence, bounding box. Steelbar, brick and pvcpipe act as proxies for structural, masonry and MEP first-fix activity in a zone; scaffold and excavator report site condition — temporary works standing, plant active.

The system is deliberately scoped as a **triage tool**. It directs a person to the photographs worth examining. It does not replace the examination, and it has no safety function of any kind. The entire pipeline runs in the browser — Roboflow for data, Google Colab for training — with no local installation, so a third party can reproduce every result from the repository alone.

## METHOD

A public dataset, *seungyeon/construction-site-km7bh* (700 images, CC BY 4.0), was forked into our own Roboflow workspace and re-split from 70/20/10 to the required 80/20. We added nothing: of 25 candidates, 20 were watermarked stock previews with no usable licence and 5 were too few to matter. Seven first-party photographs are held back as an unbiased test set.

Every label follows a written class-definition contract that specifies inclusions, exclusions, the one-box-per-visually-separable-group rule and a 20-pixel minimum size. The contract exists because without written rules, annotation drifts between people and the model learns the drift. SAM 3 and GPT-6 Astra were then run on the held-out images: both beat our own model on class accuracy, and neither produced output usable against our grouping rule.

|                |                     |
|----------------|---------------------|
| Dataset images | 700                 |
| Split          | 560 / 140 (80 / 20) |
| Model          | yolov8s.pt          |
| Epochs         | 30                  |
| Image size     | 640                 |
| Batch          | 16                  |
| Hardware       | Tesla T4, 15360 MiB |
| Training time  | 6.4 min             |

## RESULTS

| Class       | Precision | Recall | mAP@50 | mAP@50-95 |
|-------------|-----------|--------|--------|-----------|
| All classes | 0.950     | 0.884  | 0.943  | 0.809     |
| brick       | 0.984     | 1.000  | 0.995  | 0.932     |
| excavator   | 0.969     | 1.000  | 0.995  | 0.883     |
| pvcpipe     | 0.986     | 0.930  | 0.964  | 0.835     |
| scaffold    | 0.955     | 0.875  | 0.979  | 0.858     |
| steelbar    | 0.855     | 0.612  | 0.781  | 0.535     |

## INTERPRETATION

1. **Validation says 0.943 mAP@50. Our own photographs say almost nothing.** Across seven held-out images the model produced one marginal correct detection, two confident wrong-class detections and four blanks — including both classes we had designated as controls at recall 1.000.
2. **The cause is scale, and it is testable.** Holding weights, image and confidence fixed and changing only the input resolution, a blockwork wall moves from scaffold 0.78 to brick 0.46. A model whose class prediction depends on resampling has learned a texture signature, not an object. Lowering confidence to 0.02 recovers nothing.
3. **The fix is data, not capacity.** Re-split by scene group so validation can detect this at all; add wide-field imagery and train multi-scale; add hard negatives for the two confusions actually observed. A larger model would raise the leaking number and change none of the seven held-out results.

## ERROR ANALYSIS AND NEXT ITERATION

Three false positives and three false negatives are documented with filenames in *docs/error\_analysis.md*. Five of the six are explained by one cause — objects recognised at the training set's scale and not at any other — and the sixth by an annotation-convention dispute. Three prioritised data improvements follow, each naming a concrete action and the metric it should move.

## RISK: THE ASYMMETRY BETWEEN THE TWO ERRORS

**A false negative is silent.** Steelbar is placed, the model misses it, no record is created, and nobody investigates because nobody knows there is anything to investigate. The gap propagates into the programme unchallenged. **A false positive is loud.** A phantom record is contradicted by the next site walk. It costs a wasted verification trip and is self-correcting.

For progress logging, therefore, **recall is the priority**. For site-condition alerts the asymmetry reverses: repeated false alarms teach supervisors to ignore the system. **One model, two thresholds** — low for logging, high for alerting — both read off the precision-recall curve rather than left at the default.

## LIMITATIONS

- Reported metrics are inflated by train/validation leakage; the held-out set confirms it independently. They measure memorisation, not generalisation.
- Predictions change class with input resolution, so `imgsz` must be pinned end to end and confidence is not comparable across distributions.
- Wide-field shots fail — which is how site progress photography is actually taken. Validated only on close, subject-filling images.
- No safety function: no people, no PPE, no hazards. Treating it as a safety system would reduce site safety, not improve it.
- Presence only — never quantity, spacing, cover or conformance, and never a payment application. A human reviews every output.

## GOVERNANCE AND LICENSING

| Artefact             | Licence   | Owned   | Constraint                                        |
|----------------------|-----------|---------|---------------------------------------------------|
| Code and notebooks   | MIT       | Yes     | None                                              |
| Base dataset         | CC BY 4.0 | No      | Attribution; referenced, not redistributed        |
| Held-out test images | CC BY 4.0 | Yes     | Our own photographs; never trained on             |
| Trained weights      | AGPL-3.0  | Derived | Commercial use needs AGPL compliance or a licence |

No class detects or tracks people — by design: a person or PPE class would make this EU AI Act high-risk.